Fleet Overview

2025 . confirmed HDD models . reliability and deployment

Annualised Failure Rate

AFR
1.36%

Recorded Failure Flags

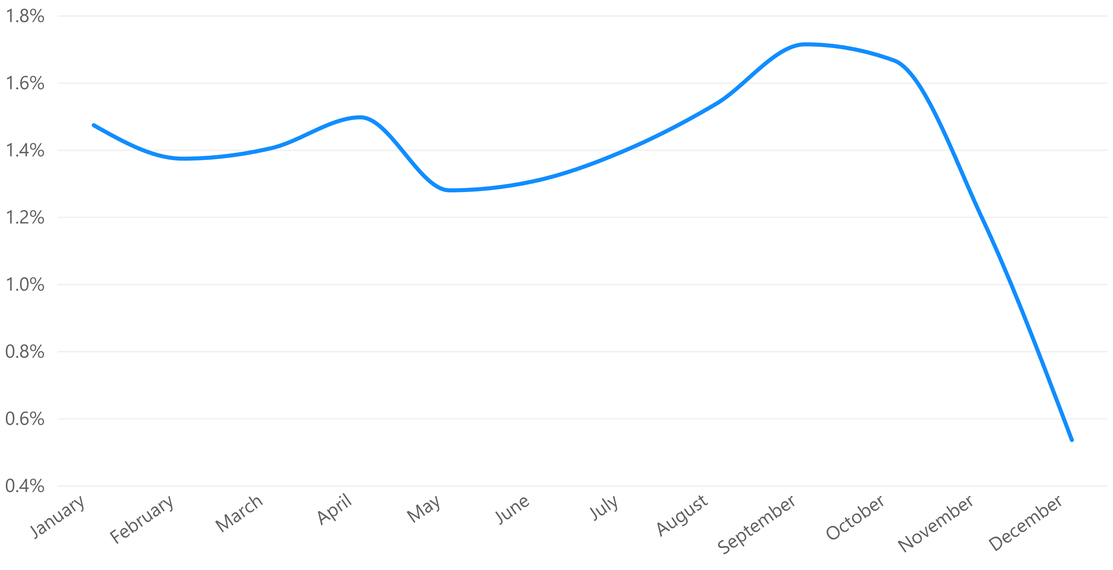
Failure Flags
4310

Top 3 models support 41% of HDD capacity

Reliability reviews should consider deployment size alongside failure patterns, because widely deployed models support more storage capacity.

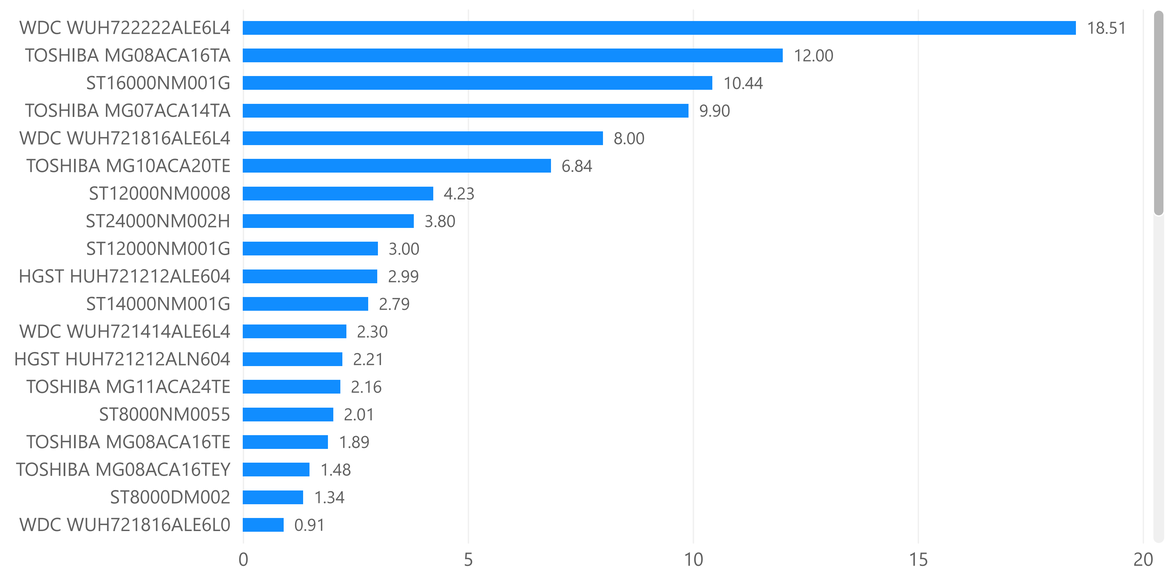
Monthly HDD Annualised Failure Rate — 2025

AFR by Month



HDD Capacity Share by Model — 31 Dec 2025 (%)

Sum of capacity_share_percent by model



How to read this page

AFR accounts for observed drive-days. Capacity share shows installed HDD capacity on 31 December 2025. Failure flags include repeated records for some drives.

Model Reliability Review

Compare failure rates, uncertainty and deployment before deciding which models need investigation.

Review status

- [ ] Does not meet the current screening .
- [ ] Investigate recurring rate pattern
- [ ] Operational-event case study

Drive capacity (TB)

0.50 — 26.00

drive model	Failure flags	Annual AFR (%)	Months above fleet AFR	Other capacity peers AFR (%)	Year-end capacity share (%)	Review status	Holm elevated	AFR lower 95% (%)	AFR upper 95% (%)
WDC WUH722222ALE6L4	176	0.47	0		18.51	Does not meet the current screening rule	False	0.41	0.55
TOSHIBA MG08ACA16TA	362	0.91	0	1.09	12.00	Does not meet the current screening rule	False	0.82	1.00
ST16000NM001G	182	0.54	0	1.23	10.44	Does not meet the current screening rule	False	0.46	0.62
TOSHIBA MG07ACA14TA	431	1.16	2	1.43	9.90	Does not meet the current screening rule	False	1.05	1.27
WDC WUH721816ALE6L4	174	0.66	0	1.14	8.00	Does not meet the current screening rule	False	0.57	0.77
TOSHIBA MG10ACA20TE	80	0.60	0		6.84	Does not meet the current screening rule	False	0.47	0.74
ST12000NM0008	485	2.58	12	2.31	4.23	Investigate recurring rate pattern	False	2.36	2.82
ST24000NM002H	146	2.76	9	0.69	3.80	Investigate recurring rate pattern	True	2.33	3.24
ST12000NM001G	124	0.94	1	2.81	3.00	Does not meet the current screening rule	False	0.78	1.12
HGST HUH721212ALE604	357	2.69	11	2.31	2.99	Investigate recurring rate pattern	False	2.42	2.99
ST14000NM001G	141	1.34	5	1.24	2.79	Does not meet the current screening rule	False	1.13	1.58
WDC WUH721414ALE6L4	70	0.82	0	1.33	2.30	Does not meet the current screening rule	False	0.64	1.03
HGST HUH721212ALN604	389	3.92	12	2.09	2.21	Investigate recurring rate pattern	True	3.54	4.33
TOSHIBA MG11ACA24TE	7	0.69	1	2.76	2.16	Does not meet the current screening rule	False	0.28	1.42
ST8000NM0055	288	2.16	11	1.87	2.01	Investigate recurring rate pattern	False	1.92	2.43
TOSHIBA MG08ACA16TE	79	1.32	4	1.01	1.89	Does not meet the current screening rule	False	1.05	1.65
TOSHIBA MG08ACA16TEY	318	6.31	10	0.79	1.48	Operational-event case study	True	5.64	7.05
ST8000DM002	155	1.73	0	2.33	1.24	Does not meet the current screening rule	False	1.47	2.02

Holm adjustment accounts for testing multiple models. True indicates an elevated rate against other same-capacity models under the test assumptions; rates are not adjusted for drive age or workload.

A high rate needs context

Read AFR alongside its confidence interval and deployment size. A high rate based on very few failures can be uncertain. Review status identifies investigation candidates; it does not establish the cause of failure.

What the colours mean

Orange: Observed in all 12 months, above the HDD fleet AFR in at least 9 months, and annual AFR above its original capacity-group benchmark.
Blue: Operational-event case study requiring separate interpretation.
Uncoloured: Does not meet this screening rule; this does not prove the model is reliable.

Toshiba: Understanding the Failure Spike

TOSHIBA MG08ACA16TEY | 2025 operational-event case study

Recorded Failure Flags

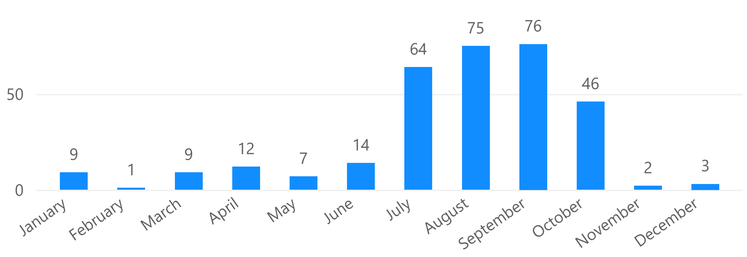
Failure Flags
318

Annualised Failure Rate

AFR
6.31%

Monthly Recorded Failure Flags — 2025

Failure Flags by Month



82% of annual flags occurred from July to October

The model recorded 261 of its 318 annual failure flags during these four months. This concentration identifies the period requiring closer investigation.

Investigation does not establish cause

Failure flags alone cannot distinguish mechanical problems from operational activity. Maintenance records and firmware context are needed to interpret the spike.

Recommended action

Review maintenance and firmware records for July–October before making replacement decisions. Track subsequent failure patterns to assess whether the elevated rate persists.

Which models need closer attention?

Investigation priorities based on recurring patterns, capacity comparisons and deployment.

Drive model	Recommended Investigation
HGST HUH721212ALE604	Review recurring flags and compare operating conditions.
HGST HUH721212ALN604	Review recurring flags, drive ages and replacement records.
ST10000NM0086	Review the high absolute rate across the 10 TB comparison group.
ST12000NM0007	Investigate recurring elevated rates and drive-age differences.
ST12000NM0008	Review recurring flags alongside its substantial deployment.
ST14000NM0138	Investigate the large difference from capacity peers.
ST24000NM002H	Check deployment growth and monthly patterns together.
ST8000NM0055	Monitor trends; annual rate is close to capacity peers.
TOSHIBA MG07ACA14TEY	Review uncertainty and age before escalating.
TOSHIBA MG08ACA16TEY	Review firmware-maintenance context; do not infer mechanical failure.
WDC WUH721816ALE6L0	Compare age and operating conditions with other 16 TB models.

Before making replacement decisions

These recommendations identify where to investigate next. Consider failure patterns, comparison evidence and deployment together. Drive age and actual workload remain unresolved, so the results do not justify automatic replacement.

How this analysis was carried out

Data coverage

The analysis uses Backblaze's 2025 daily drive records and focuses on 51 confirmed HDD models. Data is available for 364 dates; 27 September is missing. Missing observations were not treated as zero failures.

How the main measures work

Annualised failure rate (AFR): Recorded failure flags ÷ observed drive-days × 365 × 100. This accounts for observation time; it is not a prediction of future failures.

Capacity share: Each model's share of installed HDD capacity on 31 December 2025. It measures deployment, not actual storage use or workload.

Failure flags: Recorded flags can include repeated records for the same drive, so they are not always unique failed drives.

How models were selected for review

The recurring-pattern screen selected models observed in all 12 months, with AFR above the HDD fleet rate in at least 9 months and annual AFR above their capacity-group benchmark.
A separate comparison tested each model against other models of the same capacity, excluding itself. Holm adjustment accounted for testing multiple models. This comparison did not adjust for drive age or workload.

Exploratory clustering

K-means grouped 28 models using annual AFR, average daily fleet size and variation in monthly AFR. The inputs were transformed and standardised before clustering.
The main analysis produced two groups. However, the groups changed when minimum data requirements changed, so clustering was used to explore patterns rather than rank models for replacement.

What this analysis cannot establish

Drive age and actual workload were not fully accounted for. Models also differed in observation coverage and operating conditions. Rates based on very few failure flags can be highly uncertain.
Failure flags alone do not establish mechanical failure or its cause. The findings support further investigation and monitoring; they do not justify automatic replacement.

Data source
Backblaze Drive Stats — 2025 daily drive data.
https://www.backblaze.com/cloud-storage/resources/hard-drive-test-data